Repeated Parental Deprivation in the Infant Common Marmoset *(Callithrix Jacchus*, Primates) and Analysis of its Effects on Early Development

Andrea C. Dettling, Joram Feldon, and Christopher R. Pryce

Background: *We describe a successful demonstration that repeated early deprivation of parental care (ED), as used to study effects of early life stress in rats, can be performed in a primate and that it constitutes an early life stressor.*

Methods: *Seven breeding pairs of marmoset monkeys each provided control twins (CON) and twins that were subjected to ED for 30–120 min/day on postnatal days (PND) 2–28. Urine samples were obtained to monitor the acute effect of ED on cortisol and catecholamine levels. Behavior samples were obtained in the home cage to monitor the effects of ED on infant and infant–parent behavior.*

Results: *Early deprivation of parental care caused acute increases in cortisol, epinephrine, and norepinephrine. At PND 28, basal cortisol was reduced in ED compared with CON infants, and ED infants were smaller than CON infants. Early deprivation infants tended to spend more time in the suckling position than did CON. Early deprivation infants demonstrated more distress vocalization than CON infants, even though the parental care they received in the home cage was similar. Early deprivation infants tended to play less socially than did CON.*

Conclusions: *To our knowledge, this is the first demonstration that a repeated early life stressor of the type developed in rats can also be applied to a primate species.*
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Key Words: Early deprivation, nonhuman primate, cortisol, catecholamines, parental and infant behavior

Introduction

There is growing evidence that adverse early life events constitute one of the major risk factors for the development of mood and anxiety disorders in adolescence and adulthood (Bernet and Stein 1999; Glaser 2000; Heim and Nemeroff 2001; Heim et al 1997; Wong and Licinio 2001). In interplay with the genome, the early environment regulates brain development and consequently the development of species-specific behavior, as well as individual differences in behavior (Gottlieb 1998; Greenough et al 1987; Plomin 1994). In mammals, parental care is the major factor in the early environment, and there is compelling evidence that parental care commensurate with maintenance of the infant's homeostasis is important for the short-term and long-term development and functioning of neurobiology, physiology, and behavior (Hofer 1987, 1994b). This rodent and primate evidence is provided by observational studies of the consequences of spontaneous differences in maternal care and by experimental studies involving manipulation of the infant–mother dyad (Kraemer et al 1989; Liu et al 2000; Meaney et al 1996; Pryce et al 2001a; Rosenblum et al 1994). This approach in animals could yield models of the short-term and long-term consequences of abuse and neglect in human parent–infant relationships. Models of the short-term consequences of early life stress will be directly relevant to child psychiatry. Studies of the long-term consequences can provide models of important symptoms of neuropsychiatric disorders, including mood disorders and psychoses, that can (but do not have to) include adverse early life events in their etiology.

Most experimental evidence for the enduring effects of manipulation of the infant's maternal environment on its neurobehavioral development is based on rat studies. The effects of a number of different manipulation paradigms have been investigated. *Early handling* (EH) involves daily separation of pups from the mother, and in some laboratories also from the littermates, for a brief period (3–15 min), and its effects are studied typically in adulthood relative to mothers and pups that do not experience any environmental disturbance, so-called early *non-handling* (NH) (Levine 1960). *Maternal separation* (MS) involves daily separation of the intact litter from the mother for a prolonged period (3–6 hours), and *early deprivation* (ED) or *isolation* involves daily separation of pups from the mother and littermates for a prolonged period (3–6 hours). The effects of MS and ED have been studied relative to NH or to mothers and pups that

From the Laboratory for Behavioural Neurobiology, Swiss Federal Institute of Technology Zurich, Schwerzenbach, Switzerland.
Address reprint requests to Christopher R. Pryce, Ph.D., Swiss Federal Institute of Technology Zurich, Laboratory for Behavioural Neurobiology, Schorenstrasse 16, 8603 Schwerzenbach, Switzerland.
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experience the environmental disturbance, including occasional brief handling, of typical *animal facility rearing* (AFR). As adults, EH offspring demonstrate reduced stress-related neuroendocrine responses (hypothalamic–pituitary–adrenal [HPA] axis) and reduced fearfulness in comparison with NH (Caldji et al 1998; Levine 1960, 2000; Meaney et al 1996; Pryce et al 2001a). Early deprivation adult offspring exhibit reduced (i.e., EH-like) stress/fear-related endocrine and behavioral responses relative to NH (Kosten et al 2000; Pryce et al 2001a) and AFR offspring (Ogawa et al 1994).

In comparison to the large number of rat studies, the study of long-term effects of manipulation of the infant–mother dyad in nonhuman primates has, with some notable exception, received little attention. Primate studies could well provide an important complement to rodent studies in this research area. For example, the prefrontal cortex and amygdala are major brain areas in the regulation of emotion and motivation in nonhuman primates and humans, and have undergone considerable development in primates, leading to marked primate–rodent differences (Drevets 2001; Rolls 2000). Second, the infant–parent relationship differs markedly between rodents and primates: for a rat pup, it is natural to be separated from the mother (but not the littermates) for periods of 15–30 min while the mother leaves the nest for foraging. In contrast, the 1–2 primate infants are in full and continuous body contact with the mother, or in the case of some primates, mother and father, throughout the first several weeks of life (Pryce 1996). Therefore, interrupting body contact for even brief periods is likely to constitute a severe stressor for a primate infant. Rats and primates also differ in their maturational state at birth, with rats being poorly developed at birth and undergoing rapid postnatal growth, and nonhuman primate infants being born well developed and undergoing relatively slow postnatal growth. Differences also exist in the postnatal state of the HPA system. Postnatal days 2–14 constitute the stress hypo-responsive period (SHRP) of the rat HPA. Maternal care is an important regulator of the SHRP, so that postnatal manipulations such as EH and ED might exert some of their long-term effects via SHRP disruption (Levine 2000). Primate infants do not demonstrate a SHRP, with neonates already capable of adult-like pituitary–adrenal stress responses (Bowman and Wolf 1965; Gunnar 1989). In the common marmoset monkey, infants actually demonstrate high basal levels of adrenocorticotropic hormone (ACTH) and cortisol relative to those of older conspecifics (Pryce et al 2002).

The majority of primate studies have been conducted with macaques and based on *maternal privation;* that is, the complete and continuous absence of the biological mother. Following separation from the mother within the first few days after birth, infants are either nursery-reared without physical contact with any other monkey during the first 6 months of life, or peer-reared in age-matched groups of maternally deprived monkeys. In contrast to rat ED, maternal privation constitutes a chronic absence of maternal care rather than the repeated stress of temporary deprivation of such care. Maternal privation has chronic physiologic, neurochemical, and behavioral effects. In comparison to mother-reared macaques, juvenile-adolescent maternally deprived peer-reared macaques exhibited either lower basal and stress-related HPA activity (Clarke 1993) or higher basal HPA activity (Higley et al 1992), and a phase shift in the HPA circadian rhythm has also been proposed (Boyce et al 1995). Nursery-reared macaques demonstrate reduced basal cerebrospinal fluid (CSF) norepinephrine relative to mother-reared macaques (Kraemer 1992; Kraemer et al 1989). Peer-reared macaques, in contrast, demonstrate increased basal CSF norepinephrine relative to mother-reared control macaques; stress-induced levels of CSF norepinephrine and metabolites were reduced, as were basal serotonin and dopamine metabolites (Clarke et al 1996; Higley et al 1992). The chronic behavioral consequences of maternal privation include self-clutching, stereotyped body rocking, deficiencies in social behavior, and affective flattening (Gandelman 1992; Kraemer 1992; Paul et al 2000).

Primate studies that have used repeated infant–mother separations have been performed with mature, semi-independent infants, for example in 3–7-month-old squirrel monkeys (Coe et al 1983; Hennessy 1986) and 6-month-old rhesus macaques (Hinde and McGinnis 1977). These mature infants demonstrate a cortisol stress response to each separation, whereas distress vocalization decreases markedly across separations. Reunited infants spend more time in contact with the mother and less time playing. Repeated deprivation of 5 hours per week in 3–5-month-old squirrel monkeys led to increased glucocorticoid feedback sensitivity in adulthood (Lyons et al 2000).

Another approach used in primates is manipulation of maternal care received by mature infants via varying the feeding demands on the mother. When tested as young adults, bonnet macaque offspring of mothers exposed to unpredictable foraging demands exhibited elevated basal corticotropin-releasing factor levels in the CSF, increased noradrenergic and blunted serotonergic responses, and depression-like behavior, including reduced social interactions (Rosenblum and Andrews 1994; Rosenblum et al 1994). Using the correlational approach, it has been demonstrated that the primate offspring of less responsive

parents demonstrate higher stress reactivity (Dettling et al 1998; Gunnar et al 1981).

To our knowledge, there has been no attempt to date to develop a repeated early deprivation procedure, such as that conducted with rat pups, in a nonhuman primate. Here we describe a study in which we successfully conducted daily ED across the first month of life in the common marmoset (*Callithrix jacchus*). This is a small-bodied neotropical primate that in terms of reproduction is characterized by twinning and biparental care (Pryce 1993). In contrast to rat pups, marmoset infants are in continuous body contact (i.e., being carried) with a caregiver for 24 hours per day throughout the first 2–3 weeks of life (Ingram 1977; Pryce 1993), such that even a short period of ED is a nonbiological event. Further to the description and validation of ED per se, we describe some of the early effects of this manipulation in terms of stress-related endocrine responses to ED and infant–parent behavior in the family group.

Methods and Materials

Subjects and Husbandry

This study was conducted under experimental permit in accordance with the Animal Protection Act (1978), Switzerland. Seven established breeding pairs of common marmosets each contributed two sets of twin offspring, which were the subjects of this study. Each study group comprised the breeding female, breeding male, and the study twin infants only. No previous offspring were present, and the first set of twins per group was euthanized (aged 1 year) before the birth of the second set. These breeding pairs and their offspring were maintained in colony rooms each holding 2–6 such groups, with groups in auditory and olfactory but not visual or tactual contact. The home cages measured 3–4 m^3 and were equipped with natural branches and with sawdust on the cage floor. Colony rooms were illuminated by artificial lighting on a 13:11 hours light:dark cycle. Daily feeding comprised high-protein porridge containing vitamin and mineral supplements and crickets scattered in the sawdust, with commercial high-protein pellets and drinking water available ad libitum. Twice weekly, monkeys were fed fruit and egg. All offspring were born during the dark phase, and the following day was designated as postnatal day (PND) 1. The first set of twins of each breeding pair was allocated equally and at random (i.e., according to birth order) to the early deprivation (ED, $n = 7$) or the control group (CON, $n = 7$). The second set of twins of each breeding pair was assigned to the other treatment group. The seven CON litters constituted four male–female, two male–male, and one female–female twin pairs (eight male, six female monkeys), and the seven ED litters constituted six male–female and 1 male–male twin pairs (eight male, six female monkeys).

Repeated Early Deprivation

In the case of all groups, on PND 2 at 9:00 AM, the parent carrying the infants was caught and briefly restrained, the infants were removed to a procedures room and sexed, weighed, and one twin was marked by shaving some of the back and tail hairs. In the case of triplets, the least active or smallest infant was euthanized by sodium pentobarbital overdose. In the case of the seven sets of CON infants, these were returned immediately to the parents, with the time from removal to return to the parent and the home cage being approximately 5 min. Postnatal day 2 ED infants were isolated for 30 min (see below) and then returned to the parents.

On PNDs 2–28 (4 weeks), ED was conducted daily according to a fixed schedule for between 30 and 120 min per day, beginning at different time points from 8:30 AM to 5:00 PM. For each of weeks 1–4, a total of 9 hours ED was administered, comprising 2 × 30-min, 1 × 60-min, 2 × 90-min, and 2 × 120-min ED sessions. Variable durations as well as variable times of day of ED were chosen to add unpredictability to the manipulation. An additional condition of the manipulation protocol was that ED duration on PNDs 2 and 3 was set at 30 min per day. Early deprivation began with the infant being removed from the carrying parent's back while restraining the parent in the home cage. The infant was then taken to the procedures room where it was weighed. We then attempted to obtain a urine sample (0.1–0.5 mL) by gently stimulating the surface of the ano-genital region with a clean pipette tip. The infant was then placed alone in a plastic mouse cage (Macrolon, Switzerland, 25 × 19 × 14 cm) with an aluminum mesh top to which the infant could cling. This cage was placed in an isolation chamber (Coulbourn Instruments, Allentown, PA) fitted with a 4 W light. Temperature within the isolation chamber was 23–25°C. The procedure room was sound-isolated from the colony rooms. Early deprivation was conducted consecutively with each infant in each pair, so that one infant remained with the parents at all times. At the end of the ED session, we attempted to collect a post-ED urine sample using the same method. The infant was returned to the sleeping box attached to the home cage, from where it was retrieved immediately by one of the parents. In the case of CON infants, the carrying parent was restrained briefly and then released, to simulate this component of the ED procedure. On PND 28, the CON infants were removed, weighed, a urine sample was obtained via ano-genital stimulation, and the infant was then returned to the parents.

Measurement of Urinary Cortisol and Catecholamines

Urinary cortisol and catecholamine titers in samples obtained pre- (basal) and post-ED on PNDs 2–28, were measured for five of the seven sets of ED twin subjects. Basal urinary cortisol titers were measured on PND 28 for ED and CON twin subjects. Immediately after collection, infant urine samples were stored at −25°C and transferred to −80°C within 48 hours. Determination of total urinary cortisol values was performed within a single radioimmunoassay (RIA) following enzyme hydrolysis of 5 μL urine, as described in Dettling et al (1998) with the only protocol change being that ^{3}H-cortisol ([1, 2, 6, 7-^{3}H] cortisol, specific activity 80–105 CI/mmol; Amersham International, Amersham, UK) was used as tracer. Intra-assay precision was 3% ($n = 6$). As a validation step, conducted with infants that were otherwise part

of a separate study (Pryce et al 2002), matched blood and urine basal samples were obtained from eight infants aged between 2 and 28 days, one set of samples per infant. Plasma and urinary cortisol were determined in these samples to provide an estimate of the extent to which urinary cortisol values predict plasma cortisol values in this primate.

The urinary concentrations of epinephrine and norepinephrine were measured using a commercially available RIA kit for human plasma and urinary epinephrine and norepinephrine (BI-CAT-RIA, DLD Diagnostika, Hamburg, Germany; Manz et al 1990), validated in-house for marmoset urine. Before RIA, the epinephrine and norepinephrine in 10-μL urine samples were extracted using a cis-diol-specific affinity gel, and aceylated to *N*-acylepinephrine and *N*-acylnorepinephrine, respectively. Thereafter they were converted enzymatically as part of the kit procedure to *N*-acylmetaepinephrine and *N*-acylmetanorepinephrine. The RIA used rabbit antisera specific either to *N*-acylmetaepinephrine or *N*-acylmetanorepinephrine; *N*-metaepinephrine-[^{125}I]Bolton-hunter-reagent-conjugate or *N*-metanorepinephrine-[^{125}I]Bolton-hunter-reagent-conjugate, as tracers; and goat antirabbit-γ-globulin as secondary antibody. Intra-assay precision was 6% ($n = 4$) for epinephrine and 5% ($n = 4$) for norepinephrine.

Urinary concentrations of cortisol, epinephrine, and norepinephrine were expressed relative to urinary creatinine (Cr) content to control for variation in urinary volume/concentration. Creatinine was measured using a commercially available kit (Beckman Creatinine Kit 555A, Sigma-Aldrich Chemie, Schnelldorf, Germany) adapted in-house for use on microtiter plates. Intra-assay precision was 1%.

Observation of Home Cage Behavior

Behavior of subjects relative to their social and physical environments was measured in the home cage during postnatal weeks (W) 1–8. The ethogram used was based on those already published for the marmoset (Pryce et al 1995; Stevenson and Poole 1976). Observations, lasting 60 min each, were performed three times per week (3 hours/week/group) using a one-way viewing screen, with coded data entered into a handheld computer (Workabout, PSION, London, UK), running Observer mobile support package software (Noldus Information Technology, Wageningen, NL). The twin infants were the focal subjects, with behaviors recorded using all-occurrence sampling or 30-sec instantaneous sampling, and expressed as behavioral frequency/hour or behavioral duration in percent time, respectively. Relationships and behavior elements of interest, with the main examples of the latter given in parentheses, were *parent–infant* (retrieve, carry, lick ano-genital region, agonistic behavior); *infant–parent* (move around on parent, infant in suckling position, initiate carrying, terminate carrying [forced or spontaneously], proximity); *infant–infant* (social play); and *infant alone* (distress vocalization, tail-piloerection, scratch, eat, explore, solitary play). Behavioral observations were scheduled so that they were evenly distributed between AM and PM, and, for W 1–4, prior to and following the ED session on that day. In the case of observations conducted post-ED, there was a minimum interval of 1.5 hours between the end of ED and the onset of the observation.

Data Analysis

The effect of ED on infant body weight was analyzed using three-way analysis of variance (ANOVA) with within-pair, between-subject factors of infant treatment (ED vs. CON siblings) and gender (female vs. male), and a within-subject factor of age (PND 2 vs. PND 28). Effect of ED on mother and father body weights was analyzed using two-way ANOVA with within-subject factors of treatment (ED vs.CON) and infant age (PND 2 vs. PND 28). For urinary cortisol and catecholamines values, studied in ED subjects only, male and female twins were treated as independent subjects. For each hormone, the mean value of all samples obtained between PNDs 2 and 28 was calculated for each subject for each of the following: AM pre-ED, PM pre-ED, AM post-ED, PM post-ED. These values were $\log_{10}$ transformed to normalize distribution and reduce heterogeneity of variance. Circadian changes in pre-ED (i.e., basal) urinary concentrations were analyzed by means of paired *t* tests on AM versus PM values. Basal AM and PM cortisol values were significantly different, so that the effect of ED (i.e., pre-ED vs. post-ED) on cortisol values was analyzed by paired *t* tests for AM and PM samples separately. Pre-ED catecholamine values did not differ significantly between AM and PM, so that an overall mean value was calculated per subject for analysis of pre-ED versus post-ED values using paired *t* tests. Basal AM urinary cortisol titers at PND 28 were compared in ED and CON siblings using a paired *t* test.

In the case of behavioral analysis, individual infant subjects could often not be identified with high reliability during observations; therefore, mean values were calculated for each twin pair and used for statistical analyses. Behavioral scores were analyzed using two-way ANOVA with a within-pair, between-subject factor of treatment (ED vs. CON siblings) and within-subject factor of age (W 1–4 vs. W 5–8). Behavior elements that were observed during a single age period only (for all elements where this was the case they were observed during W 5–8 only), were analyzed by means of paired *t* tests. Paired *t* tests were also used for a posteriori testing of significant two-way interactions. The significance level was set at $p < .05$, with $p < .10 > 0.05$ considered as a trend to a significant effect.

Results

Infants vocalized considerably during ED, primarily with the distress vocalization that they emit in the home cage in response to parental aggression and also with the long call that is emitted by all age groups in this species and functions to advertise location. We had the clear impression that the amount of time that infants spent vocalizing during ED decreased with age and repeated ED exposure. Early deprivation infant vocalizations were not audible to marmosets in the main colony rooms, including parents, and vice versa. Older infants also urinated and defecated during ED, in particular during ED sessions of longer duration (90–120 min). Piloerection of the body hair was

Table 1. Effect of Repeated Early Deprivation on Infant Body Weight (Grams; Mean ± SEM) at PND 2 and PND 28 Compared with Control Siblings

	ED	CON	*p* value
PND 2	31.4 ± .6	32.1 ± .7	ns
PND 28	55.5 ± 1.5	62.9 ± 2.6	< .003

The average weight of each pair of twins was used, with $n = 7$ for CON and ED.

PND, postnatal day; ED, early deprivation; CON, control.

also a frequent response to ED. All of these observations were qualitative.

Effects of Early Deprivation on Body Weight

Male and female infants did not differ significantly in body weight at either PND 2 or PND 28 as revealed by the absence of both a significant main effect of gender and interaction of age and gender ($p > 0.18$). Consequently, male–female twin weights were averaged, and these data were analyzed by two-way ANOVA. As given in Table 1, ED resulted in significantly lower body weight at PND 28 relative to CON siblings. This was supported by the significant interaction between age and treatment [$F(1,6) = 20.6$, $p < .004$] and the main effect of treatment [$F(1,6) = 21.9$, $p < .004$]. At PND 2 there was no significant difference in body weight between ED and CON infants [$t(6) = 1.01$, $p > 0.35$], whereas at PND 28, the last day of ED, ED infants were significantly smaller than CON infants [$t(6) = 5.00$, $p < .003$]. The reduction in body weight of PND-28 ED infants relative to their CON siblings was 12% on average. The average ED duration per day was 1.3 hours, and the average percent time spent by CON infants in the suckling position was 13% (see below; Table 2), so that ED infants were deprived of about 0.2 hours of suckling per day, or about 1% of the day.

Parental body weights were not affected significantly by the ED manipulation, as revealed by the absence of a main effect of treatment and age × treatment interaction, in the case of mothers ($p > .18$) and fathers ($p > .12$). At PND 2, maternal body weight was 440 ± 10 g for CON and 420 ± 16 g for ED, and at PND 28, 442 ± 11 g for CON and 437 ± 16 g for ED. At PND 2, paternal body weight was 415 ± 20 g for CON and 400 ± 19 g for ED, and at PND 28, 396 ± 25 g for CON and 400 ± 20 g for ED.

Urinary Cortisol and Catecholamine Responses to Early Deprivation

Figure 1A presents the findings for urinary cortisol measures across PNDs 2–28 in ED infants. Basal am cortisol/Cr titers were significantly higher than basal PM cortisol/Cr titers, as indicated by a paired *t* test [$t(9) = 3.79$, $p < .005$]. When ED was performed PM then infants demonstrated a significant post-ED versus pre-ED increase in their urinary cortisol/Cr titers [$t(9) = -2.81$, $p < .03$]. This was not the case when ED was performed am; that is, when basal titers were high [$t(9) = 0.19$, $p > 0.84$]. Figure 1B depicts the relationship between am plasma and urinary cortisol titers in normally reared marmoset infants: urinary cortisol/Cr values correlated significantly with the cortisol titers in matched plasma samples [$r = 0.90$; $F(1,6) = 25.95$, $p < .003$]. In the case of basal am cortisol/Cr titers at PND 28, these were significantly reduced in ED compared with CON subjects [$t(6) = -3.81$, $p < .01$, Figure 1C]. For urinary basal catecholamine titers, there was no significant circadian change for either epinephrine [$t(6) = 0.66$, $p > .54$] or norepinephrine [$t(6) = 0.82$,

Table 2. Effects of Repeated Early Deprivation (ED) on Home Cage Behavior at Ages 1–4 and 5–8 Weeks (W) Compared with Control Siblings

	ED		CON		
Behavior	W1–4	W5–8	W1–4	W5–8	*p* value
Suckling position[a]	18.2 ± 2.8	6.7 ± 1.2	13.0 ± 1.6	7.0 ± 1.3	$p < .07$[c]
Carrying by mother[a]	33.2 ± 6.9	18.4 ± 5.5	28.8 ± 5.2	14.6 ± 2.1	ns
Carrying by father[a]	63.9 ± 7.7	31.6 ± 4.7	64.7 ± 6.9	30.2 ± 6.3	ns
Ano-genital licking[b]	1.2 ± .1	.6 ± .07	1.2 ± .2	.6 ± .1	ns
Parental aggression[b]	1.6 ± .4	1.7 ± .3	1.4 ± .2	2.4 ± .5	ns
Proximity to parent[a]	.6 ± .3	10.9 ± 2.9	.9 ± .2	12.0 ± 2.0	ns
Proximity to twin[a]	.5 ± .3	12.1 ± 2.1	1.1 ± .4	16.6 ± 3.2	ns
Distress vocalisation[a]	2.5 ± .7	3.1 ± 1.2	1.6 ± .3	2.3 ± .9	$p < .05$[d]
Tail hair piloerection[a]	.05 ± .04	11.1 ± 4.0	.8 ± .4	5.1 ± 1.1	ns
Social play[a]	.04 ± .04	4.3 ± .6	.1 ± .06	5.8 ± 1.4	$p = .10$[e]
Solitary play[a]	.0 ± .0	3.9 ± .6	.02 ± .02	2.8 ± .4	

[a]Percent observation time.
[b]Frequency of occurrence per hour.
[c]Age × early condition interaction.
[d]Main effect of early condition.
[e]Play type × early condition interaction at W 5–8.

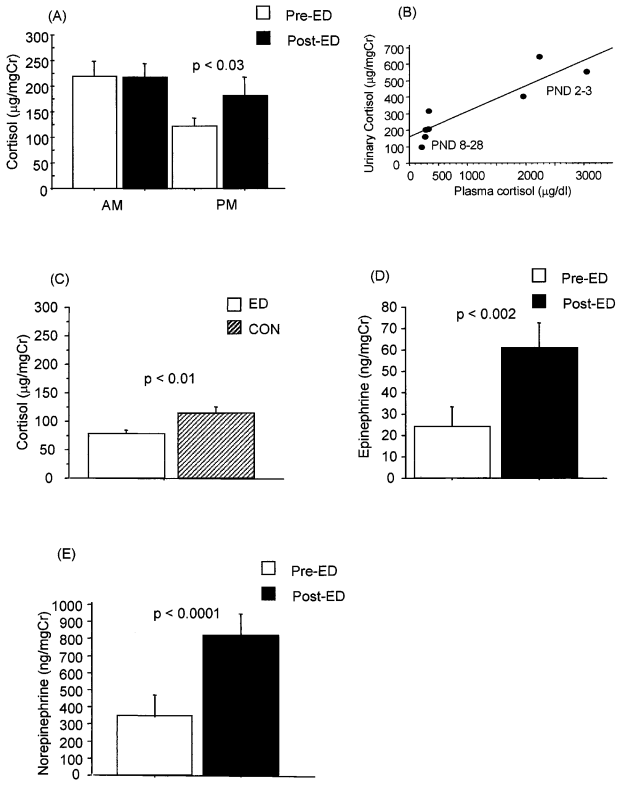


Figure 1. Marmoset repeated early deprivation and stress endocrinology. **(A)** Urinary cortisol response to early deprivation performed in the morning or afternoon on postnatal days (PNDs) 2–28. $n = 10$ early deprivation (ED) infants, with urine samples pooled, AM or PM, for PNDs 2–28 per subject for pre-ED (basal) and post-ED (90- and 120-min deprivations). **(B)** Relationship between cortisol titers in matched blood and urine samples from 8 infant marmosets aged between 2 and 28 days that were otherwise not part of this study. The solid line is for the regression equation [$Y = 0.153X + 159.7$; $r = 0.90$; $F(1,6) = 25.95$, $p < .003$]. **(C)** Morning basal (peak) urinary cortisol titers in ED infants compared with control (CON) siblings at PND 28. Note that the CON values are similar to those of the older infants depicted in B. **(D)** Urinary epinephrine response to early deprivation performed on PNDs 2–28. $n = 10$ ED infants, with urine samples pooled, AM and PM, for PNDs 2–28 per subject for pre-ED (basal) and post-ED (90- and 120-min deprivations). **(E)** Urinary norepinephrine response to early deprivation performed on PNDs 2–28. $n = 10$ ED infants, with urine samples pooled, AM and PM, for PNDs 2–28 per subject for pre-ED (basal) and post-ED (90- and 120-min deprivations).

$p > .44$]. Accordingly, am and PM values were collapsed for further analysis. Infants responded to ED with a significant increase in epinephrine [$t(9) = -5.86$, $p < .002$, Figure 1D] and norepinephrine [$t(9) = -7.49$, $p < .0001$, Figure 1E].

Early Deprivation and Behavior

All of the behavioral elements that occurred in both W 1–4 and W 5–8 (e.g., carrying, infant in suckling position, distress vocalization, play) demonstrated at least a trend to a significant main effect of age, reflecting maturation and development across weeks 1–8 of life in the marmoset. As given in Table 2, ED did not affect the percent time that subjects were carried by mothers and fathers, the frequency of ano-genital licking by mothers and fathers, the percent time subjects spent in proximity of the parents, or the frequency of parent–infant agonistic behavior, such as aggression. The absence of such effects was reflected by the lack of (a trend to) either a significant main effect of treatment or an age × treatment interaction. Early deprivation did yield the following behavioral effects: when being carried by the mother, ED infants aged W 1–4 tended to spend more time in the suckling position than did their CON siblings; this was supported by a trend to a significant age × treatment interaction [$F(1,6) = 5.12$, $p < .07$] (Table 2). Early deprivation subjects spent more time emitting distress vocalizations than did their CON siblings, as revealed by the significant main effect of treatment [$F(1,6) = 6.14$, $p < .05$, Table 2]. Distress vocalizations are typically emitted in response to the parents' agonistic attempts to terminate carrying and also when the infant is alone on the substrate. Therefore what was particularly noteworthy here was that ED infants distress-vocalized significantly more than CON infants, although they were neither aggressed against more nor carried less (Table 2). At W 5–8, play behavior was observed in the form of both solitary and social play. ED infants spent relatively more time in solitary play and less time in social play than their CON siblings, as revealed by the trend to a significant interaction between play type and treatment [$F(1,6) = 3.71$, $p = .10$, Table 2].

Finally, all of the subjects included in this study survived to the end of the first year of life (young adulthood). During this time the effects of repeated ED were studied in terms of endocrine and cardiovascular parameters and spontaneous and conditioned behavior. At age 1 year, subjects were euthanized for the study of the neurobiological effects of repeated ED in this primate. These effects will be reported on elsewhere (e.g., Dettling et al, in press).

Discussion

The present study has demonstrated that repeated early deprivation, a postnatal environmental manipulation developed in the rat, can be applied to at least one nonhuman primate species, the common marmoset monkey. Primate studies based on repeated ED will represent an important complement to the rat ED studies, for investigation into

the short-term and long-term effects of continuous disruption of the early infant–parent relationship on neurobehavioral maturation and development. Given the epidemiologic evidence that early life stress, including neglect and abuse, represents an important short-term risk factor for infant physical and mental well-being, and an important long-term risk factor for the development of psychiatric disease, most notably mood disorders, then the availability of both rodent and primate repeated ED models represents an important advance in preclinical research in this area.

Using a counterbalanced crossover design, in which the same breeding pairs each contributed an ED and a control set of twin offspring, enabled us to control for some of the genetic and environmental variation between groups—most notably the inter-group differences in parental care, which are marked in the common marmoset (Pryce et al 1995)—and thereby increased the statistical power of the research design. Of course, this necessitated that the breeding pairs used remained in a stable physical and reproductive condition across the entire 2-year period for which each was studied. Before discussing the effects of ED on infants; it is important to note therefore that, based on the measure of body weight, mothers and fathers appeared to maintain a stable physical condition throughout the study period. Furthermore, these marmoset parents demonstrated behavior that was commensurate with survival of their ED infants; they demonstrated immediate retrieval of the infants at reunion following ED, and it was this retrieval that enabled the infant to cling onto the parent and initiate a carrying bout.

All of the marmoset infants survived the repeated ED procedure used in this study. Our procedure did include a process of selection in that, in the case of triplets, the smallest/weakest triplet was excluded from the study. Of course the demonstration that infants and parents can tolerate the ED procedure is a basic but essential validation of this original approach to inducing early life stress in a nonhuman primate. Repeated ED had resulted in a robust reduction in body weight by PND 28, the final day of the early life manipulation procedure. This is in contrast to a recent study in the rat in which we found that repeated ED for 4 hours at PNDs 1–21 did not affect pup body weight at PND 21 relative to control rats (Pryce et al 2001b). Smaller-than-expected body size is a well-recognized phenomenon in pediatric medicine, and early life stress might be important in its etiology (Boddy et al 2000). There are several potential explanations for this retardation in growth related to repeated ED in the marmoset, including the following: loss of homeostasis in the form of ED-induced stimulation of catabolic processes in infants; the reduction in the time available for suckling; and the reduction in milk quantity/quality due to stress experienced by ED mothers. With regard to reduced suckling time mediating the effect, as detailed in the Results, the reduction in body weight was markedly greater than can be accounted for by deprivation of suckling time per se. We were careful to distribute behavior observations throughout the day and to avoid observation for at least 1.5 hours following infant–parent reunion. Despite this, ED infants tended to be in the suckling position more than were CON infants, indicating that ED infants were able to compensate—in some cases possibly even over-compensate—for the deprivation of suckling opportunity. In the rat we observed that 4-hour ED pup–dam reunion is always followed by a bout of maternal behavior but then a rapid return to control levels of maternal care (Pryce et al 2001b). Of course, if differences in milk quality were responsible for the retarded growth in ED marmoset infants, then time spent in the suckling position would provide only a weak marker of nutritional intake, and analysis of the effects of ED on milk quantity/quality will be an important validation step; however, if stability of maternal body weight serves as a good predictor of stability of lactation, then we would not anticipate finding a marked impact of ED at the level of nutrition.

In contrast to the evidence against compromised milk intake being the mediator of the reduced weight gain in ED infants, the analysis of urinary levels of stress hormones provides supportive evidence for the hypothesis that stress-induced catabolic processes made an important contribution to this ED effect. We have recently described that infancy in the common marmoset is characterized by basal hyperactivity of the pituitary–adrenal system, with plasma basal levels of both ACTH and cortisol being markedly higher in weeks 1–4 of life than at older life stages including adulthood (Pryce et al 2002). Therefore, in terms of absolute basal values at least, this nonhuman primate represents the opposite extreme to the rat with its infantile HPA SHRP (Levine 2000). Despite the SHRP, the rat pup HPA system is sensitive to repeated ED: using a 1-hour ED at PNDs 2–9 resulted in no effect on basal plasma corticosterone levels but a marked potentiation of the plasma corticosterone response to the ED on PND 9 (McCormick et al 1998). In the present study, basal urinary cortisol levels reflected the high plasma cortisol levels of young infancy in this primate species. A cortisol response to ED was obtained in the afternoon and not in the morning; there was a circadian change in urinary cortisol levels, and the lack of a morning urinary cortisol stress response may be attributable to a remarkable ceiling effect of morning adrenocortical function in young marmoset infants. By PND 28, basal am urinary cortisol levels were reduced in ED infants compared with their CON siblings. This provides a very interesting and important parallel to the human evidence that neglected children

with psychopathology exhibit low peak basal cortisol levels and that children with PTSD symptoms exhibit enhanced dexamethasone suppression of cortisol (Heim and Nemeroff 2001). Repeated maternal separation performed with older infants in the closely related squirrel monkey consistently induced a plasma cortisol stress response but did not alter basal cortisol levels or the magnitude of the stress response (Hennessy 1986); it did however lead to increased glucocorticoid feedback sensitivity in adulthood (Lyons et al 2000).

In the rat, ED has been well studied in terms of its acute and chronic effects on HPA activity, whereas the sympathetic autonomic nervous system (ANS) has received relatively scant attention. This is despite the important role attributed to the ANS in influential models of the interrelationships between maternal care, infant physiology, and long-term development (e.g., Hofer 1994a). In the present study we were able to detect marked acute increases in the urinary levels of epinephrine and norepinephrine in response to ED. These increases provide robust quantitative evidence that repeated ED constitutes a repeated stress procedure in the infant marmoset monkey. Like the cortisol released from the adrenal cortex, epinephrine released from the adrenal medulla functions to promote conversion of nutrient reserves into glucose for energy synthesis. The ED-induced elevation of urinary epinephrine provides indirect evidence for activation of the sympathetic ANS, with the epinephrine conveyed in the blood stream to amplify the stress-induced neural activation of the sympathetic target organs. Norepinephrine is the major neurotransmitter at sympathetic ANS synapses and urinary norepinephrine, at least in the human case, is derived primarily from this source (Maas et al 1987). Therefore, our evidence suggests that repeated ED acts to activate the sympathetic ANS of infant marmoset monkeys. In rhesus macaques, maternal privation results in chronic alteration of basal and stress-related CSF norepinephrine activity (Clarke et al 1996; Higley et al 1992; Kraemer 1992). We are currently investigating the chronic effects of repeated ED in the subjects described in this study in terms of basal urinary, plasma, and CSF levels of pituitary–adrenal hormones and catecholamines.

The study of the short-term effects of the repeated ED procedure on behavior within the infant–mother and infant–father dyads is of course extremely complex: inherent to ED are periods of deprivation of parental care, deprivation of the opportunity to provide care, as well as, following reunion, altered status of the infant stimuli that interact with parental motivational factors to elicit parental care. Despite this dynamic series of consequences of ED, parental care was apparently quite unresponsive to ED. The amount of time spent carrying infants and the distribution of carrying between mothers and fathers was unaffected by ED compared with CON. Ano-genital licking of and aggression toward infants were also unaffected by ED. In rats, EH increases licking and arched-back nursing throughout the day relative to NH and AFR, whereas 4-hour ED in rats does not have this effect (Liu et al 1997; Pryce et al 2001b). Although we did not include a completely undisturbed parent–infant treatment in this study, the levels of all these parental behaviors in both CON and ED groups were very similar to those reported in previous studies conducted with undisturbed family groups (e.g., Ingram 1977; Pryce 1993).

ED infants did tend to spend more time in the suckling position on the mother, and this might have been motivated by either increased hunger or, indeed, increased comfort seeking relative to CON infants. Furthermore, ED infants aged 1–4 and 5–8 weeks spent more time emitting distress vocalization even though they received similar levels of parental care to CON infants. This might indicate that the same amount of parental care was less effective in ameliorating distress in ED infants. Finally, in terms of behavior, the data suggest that ED infants spent relatively less of their total play time in social play and relatively more in solitary play, than CON. We are continuing to observe social behavior as the subjects mature, when we would normally expect their levels of social twin play behavior to increase markedly. A relative reduction in social play as a consequence of ED would provide a very interesting measure of a long-term effect of ED on emotional function in social situations.

It is important to comment on and compare the ED control group in rodent and primate studies. In the present study the control procedure comprised brief restraint of the carrying parent and infant, and there can be little doubt that this is the most appropriate control for ED in the marmoset. In the case of rat postnatal manipulations, however, the form that the control group should take is a matter of considerable debate and controversy (Lehmann and Feldon 2000). Non-handling is widely accepted as the control group for EH, although it does not allow for the separation of the effects of handling per se versus the effects of the brief separation. Non-handling has been used as the control condition for 3-hour repeated MS (e.g., Plotsky and Meaney 1993) and has revealed that MS is without effect on a large number of behavioral and neurobiological parameters (e.g., Caldji et al 2000; Plotsky and Meaney 1993). Perhaps related to this, some recent studies have abandoned the use of NH and have moved to AFR as the control condition for MS (Huot et al 2001). Our recent rat studies have included NH and AFR as control groups and found important differences as well marked similarities between the two, depending on the parameter under investigation (e.g., Lehmann et al 2002; Pryce et al 2001a, 2001b). There is no *a priori* reason why

the brief restraint and handling of parent and infant, as performed here for the marmoset, should not also be used as a control condition for ED and other manipulations in the rat. Furthermore, there is no *a priori* reason why the crossover design used here should not also be used in rat studies.

In summary, with the present study we have demonstrated that the repeated ED can be performed successfully in a species of nonhuman primate, the common marmoset, thereby allowing for the study of the neurobehavioral effects of this early life stressor in both rodents and primates. That repeated ED does indeed constitute an early life stressor is demonstrated by its acute effects on growth and catabolic stress hormone levels, and by it leading to reduced basal urinary cortisol levels by the end of the ED protocol at PND 28. Behavioral effects of ED were observed, including at weeks 5–8 of life, that is, after the stressor had been terminated. The chronic physiologic, behavioral, and neurobiological effects of ED in the marmoset will now be studied, with the ultimate aim of establishing the suitability of this manipulation as a primate model of human early life stress and developmental psychopathology.

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